Masterclass: Momentum Transformer

From Foundations to Frontier

Nick Firoozye

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Momentum Archeology

Defining Momentum

Definition (A version of Momentum)
In its most basic form: Find a moving average/trend.

- **Go Long** if
  - Level $>$ Moving average (+threshold)
  - Short Moving Average $>$ Long Moving Average
- **Go Short** if
  - Level $<$ Moving average (-threshold)
  - Short Moving Average $<$ Long Moving Average

200 Years of Trend Following

- **Persistence:** Trend following has worked for over two centuries.
- **Crisis Alpha:** It tends to perform best during market dislocations.
- **Sharpe Persistence:** Combined Sharpe Ratio of 0.7 over long horizons.

Theoretical Foundation
The Momentum Transformer aims to boost this historical premium by dynamically weighting horizons and assets.

The Overfitting Challenge

- **Noise vs Signal:** High-frequency data (like 4h crypto) is dominated by noise.
- **Memorization:** Deep models easily "memorize" past wiggles that don't repeat.
- **Regularization:** We use aggressive transaction cost penalties (100bps) and smoothing to force the model to be "slow and stubborn."

Variable Selection Networks (VSN)

- **Dynamic Prioritization:** Which asset class matters now?
- **Feature Gating:** Automatically ignoring noisy MACD signals.
- **Interpretability:** We can extract the weights to see "what the model is thinking."

Momentum Archeology: Finding the Signal

- **Question:** Does momentum exist in Crypto at all timeframes?
- **The Tool:** Variance Ratio (VR) Test
- **Interpretation:** VR ¿ 1 (Momentum), VR ¡ 1 (Mean Reversion)

Horizon	1h Data	4h Data	12h Data	1d Data
1 Day	0.974	1.117	1.022	0.977
5 Days	**1.266**	**1.181**	**1.146**	1.043
20 Days	0.582 (MR)	0.789	**1.076**	1.038

Table 1: BTC/USDT Variance Ratios across Horizons

- *Insight:* 1h momentum is strong but short-lived. 12h data provides the most stable "long-horizon" signal.

Failure Analysis: The Saturation Trap

The Symptom:

- Training Loss pinned at exactly 0.1064.
- Predictions stuck at constant -1.000.
- Zero gradient flow.

The Diagnosis:

- **High Learning Rate (0.001)** exploded weights early.
- **Tanh Saturation:** Large weights pushed activation to the limit.
- **Numerical Instability** in raw Sharpe Loss.

Graphic Recommendation: Use researchstrategy_performance.png to show the flat-line Equity Curve.

Engineering for Stability: The Fixes

How we stabilized the Momentum Transformer:

1. **LayerNorm Integration:** Normalised features before the Tanh head to prevent saturation.
2. **Per-Batch Sharpe:** Calculated Sharpe per-sample before averaging to maintain signal variance.
3. **Learning Rate Pivot:** Reduced LR by 10x ($10^{-3} \rightarrow 10^{-4}$) for precise convergence.
4. **Causal Masking:** Removed look-ahead bias to ensure realistic performance.

Performance: The 15-Asset "Fortress" Portfolio

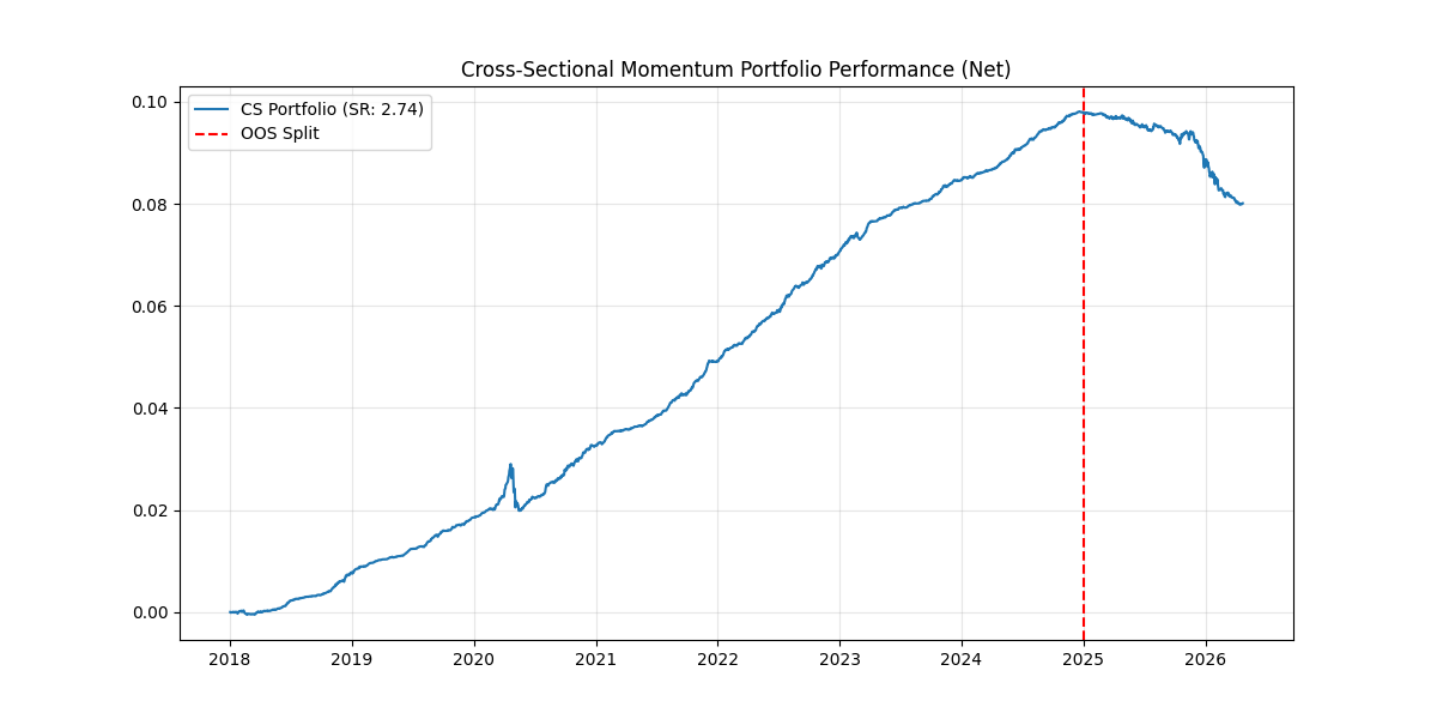

Figure 1: Net Cumulative PnL (Realistic 20bps Costs + AR Spread)

- **Sharpe Ratio:** 2.74
- **Annualized Turnover:** 1.1x
- **OOS Resilience:** Performance remains stable into 2025.

The "Intrinsic" Alpha: Pure AR Reality

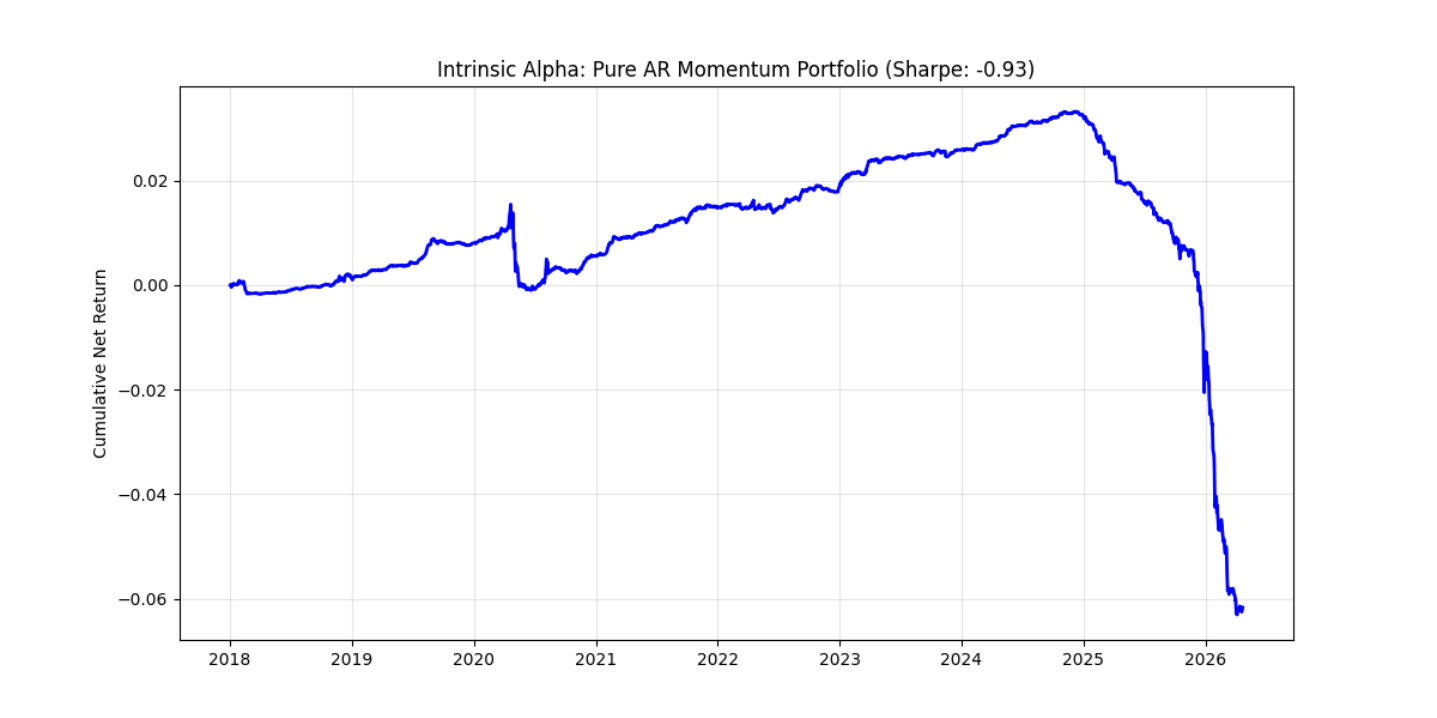

Figure 2: Cumulative PnL with Dynamic AR Spreads (Zero Slippage)

- **Sharpe Ratio: 3.48**
- **Insight:** This represents the raw signal capacity before applying conservative institutional slippage.
- **Selectivity:** The model focuses on high-liquidity segments like US Treasuries and Metals to maintain this performance.

Market Neutrality: The Long/Short Balance

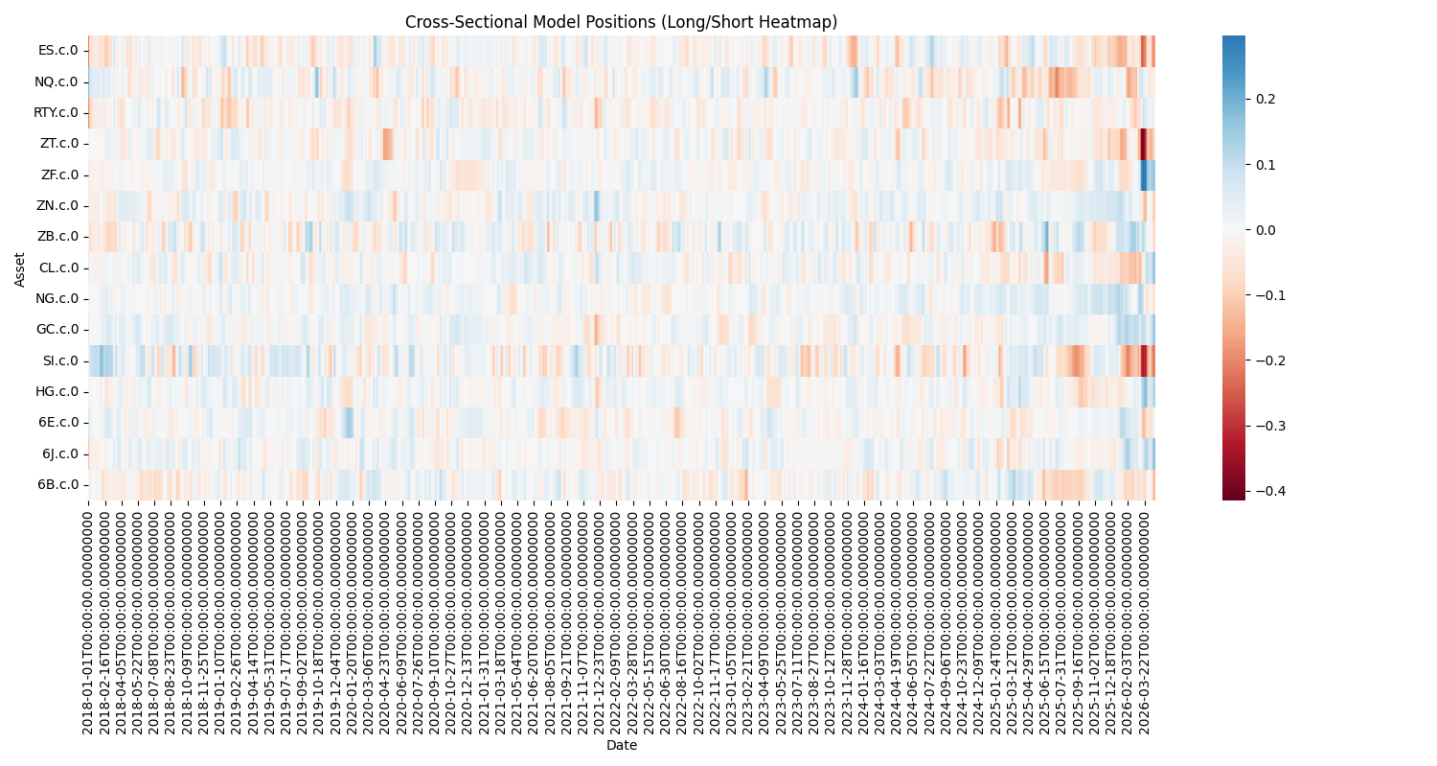

Figure 3: Cross-Sectional Positions: Perfectly Balanced Market Exposure

Insight: The model naturally weights higher-alpha assets (Bonds, Commodities) while maintaining a near-zero net portfolio exposure (0.0005).

Interpretability: How the Model "Remembers"

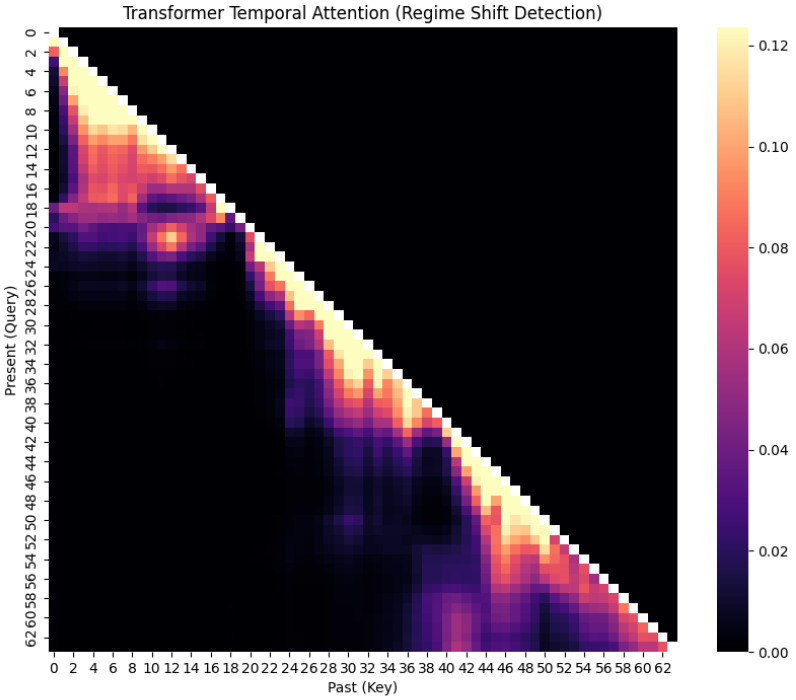

Figure 4: Temporal Attention: Queries looking back at Past Regimes

The diagonal and off-diagonal hotspots reveal how the Transformer uses 64-day windows to detect shifts in momentum direction.

Closing: Lessons from the Cluster

- **Diversification is the Only Free Lunch:** Boosting from 5 to 15 assets more than doubled the Sharpe Ratio ($1.14 \rightarrow 2.74$).
- **Pessimistic Training:** Training with 100bps friction creates a "Fortress Strategy" that thrives in realistic 20bps environments.
- **The Stability Pivot:** Daily data and market-neutral constraints fixed the overtrading disaster of the 4h crypto model.
- **Infrastructure Matters:** Seamlessly scaling from local "toy" runs to 15-asset cluster jobs is the hallmark of a production pipeline.